

Supporting Information

Highly active and controllable MOF-derived carbon nanosheets supported iron catalysts for Fischer-Tropsch synthesis

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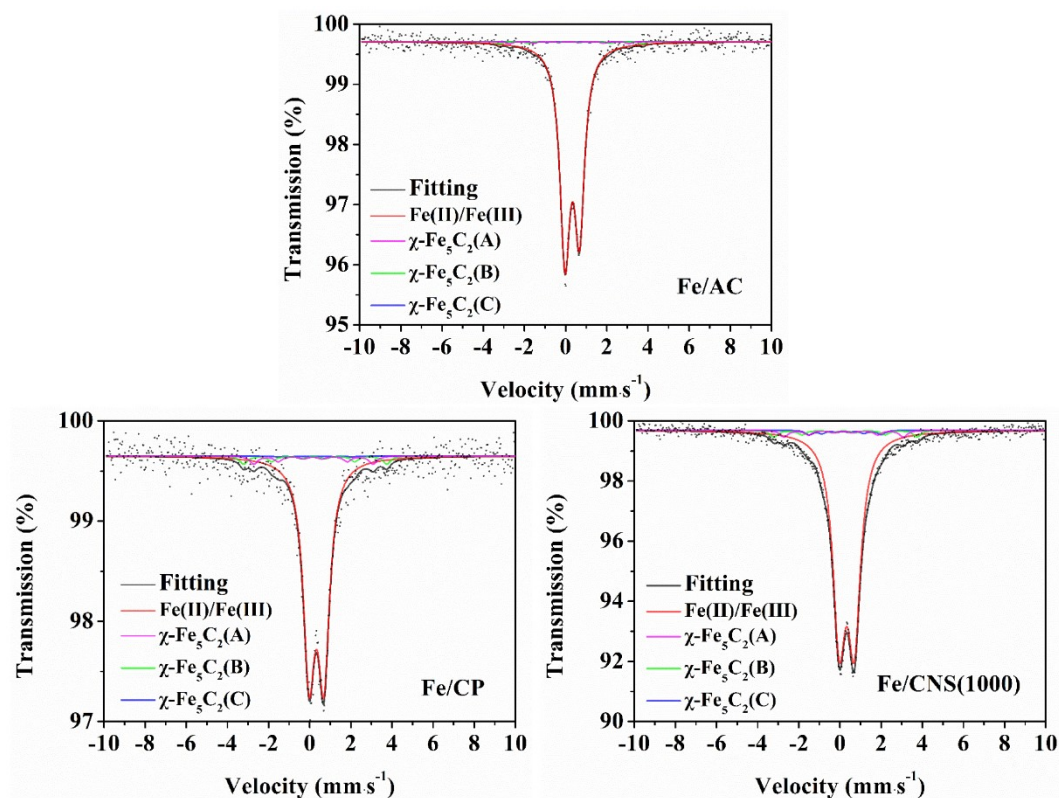


Fig. S1. ^{57}Fe Mössbauer spectra of the fresh Fe/AC, Fe/CP and Fe/CNS(1000) catalysts.

Table S1 ^{57}Fe Mössbauer parameters of the fresh Fe/AC, Fe/CP and Fe/CNS(1000) catalysts.

Fresh catalysts	IS ($\text{mm}\cdot\text{s}^{-1}$)	QS ($\text{mm}\cdot\text{s}^{-1}$)	H (T)	Phase ascription	A (%)
Fe/AC	0.32	0.70	-	Fe(II)/Fe(III)	97.2
	0.17	0.03	18.0	$\chi\text{-Fe}_5\text{C}_2(\text{A})$	1.1
	0.22	0.09	21.6	$\chi\text{-Fe}_5\text{C}_2(\text{B})$	1.1
	0.20	0.11	11.1	$\chi\text{-Fe}_5\text{C}_2(\text{C})$	0.6
Fe/CP	0.33	0.72	-	Fe(II)/Fe(III)	89.8
	0.17	0.03	18	$\chi\text{-Fe}_5\text{C}_2(\text{A})$	4.8
	0.22	0.09	21.6	$\chi\text{-Fe}_5\text{C}_2(\text{B})$	4.8
	0.20	0.11	11.1	$\chi\text{-Fe}_5\text{C}_2(\text{C})$	0.6
Fe/CNS(1000)	0.33	0.75	-	Fe(II)/Fe(III)	90.5
	0.17	0.03	18.0	$\chi\text{-Fe}_5\text{C}_2(\text{A})$	3.8
	0.22	0.09	21.6	$\chi\text{-Fe}_5\text{C}_2(\text{B})$	3.8
	0.20	0.11	10.9	$\chi\text{-Fe}_5\text{C}_2(\text{C})$	1.9

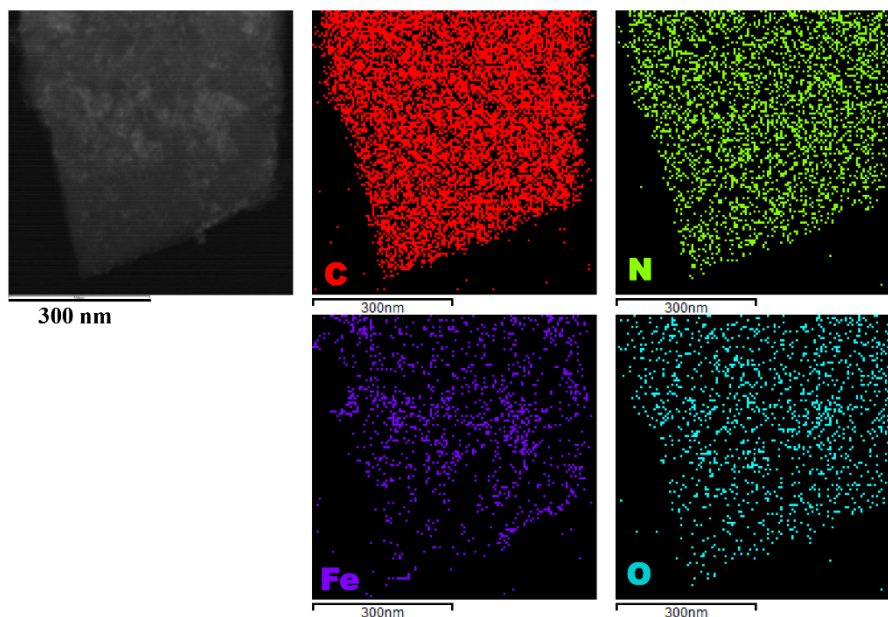


Fig. S2. TEM images of the Fe/CNS(1000) catalysts in dark field, and corresponding mappings of C, N, Fe and O.

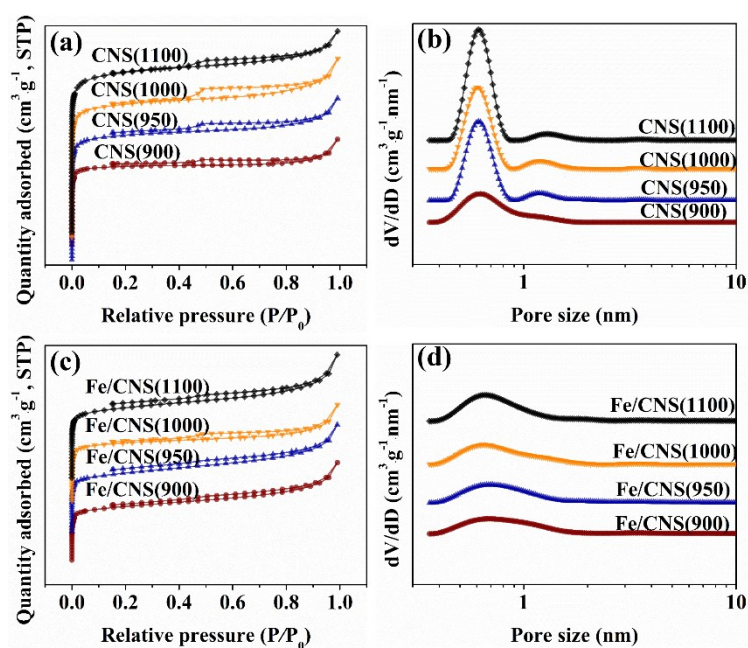


Fig. S3. N₂ adsorption-desorption isotherms (a, c) and pore size distribution (b, d) of the CNS(T), and corresponding iron-based catalysts.

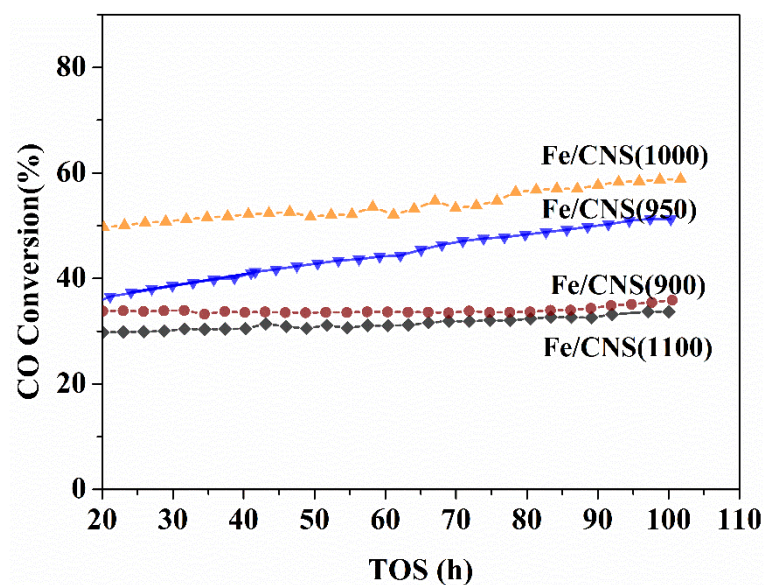
Table S2 Surface area and pore structure parameters of CNS(T) and Fe/CNS(T) catalysts.

samples	S^{BET} (m^2g^{-1})	V_{total} (cm^3g^{-1})	V_{micro} (cm^3g^{-1})	D_a (nm)
CNS(900)	272	0.12	0.10	1.7
CNS(950)	416	0.18	0.15	1.8
CNS(1000)	438	0.20	0.15	1.8
CNS(1100)	524	0.23	0.18	1.8
Fe/CNS(900)	194	0.12	0.06	2.3
Fe/CNS(950)	208	0.13	0.07	2.4
Fe/CNS(1000)	210	0.14	0.08	2.1
Fe/CNS(1100)	250	0.15	0.09	2.3

Table S3 FTS performance of Fe/CNS(T) catalysts at changeable space velocity.

Catalysts	CO conv. (%)	FTY ($\mu\text{mol}_{\text{CO}}$ $\text{g}_{\text{Fe}}^{-1}\text{s}^{-1}$)	CO ₂ selectivity (%)	Hydrocarbon selectivity (C-mol%)					O/P
				CH ₄	C ₂ -C ₄ ⁼	C ₂ -C ₄ ^o	C ₅ ⁺	Oxy.	
Fe/CNS(900)	33.9	198	31.9	26.5	28.4	15.8	28.4	0.9	1.8
Fe/CNS(950)	34.9	278	30.5	27.2	26.3	15.3	30.4	0.8	1.7
Fe/CNS(1000)	34.7	323	31.6	27.5	24.9	15.6	31.2	0.8	1.6
Fe/CNS(1100)	31.2	191	28.6	28.4	23.0	16.3	31.6	0.7	1.4

Reaction conditions: 0.1 g cat., 340 °C, 1 MPa, H₂/CO=1, GHSV=9000-14000 mL·g⁻¹·h⁻¹, TOS=30 h.

**Fig. S4.** CO conversion of Fe/CNS(T) catalysts on TOS. ($P = 1$ MPa, $\text{H}_2/\text{CO} = 1$, $\text{GHSV} = 9000 \text{ mL}\cdot\text{g}^{-1}\cdot\text{h}^{-1}$, $T = 340$ °C)

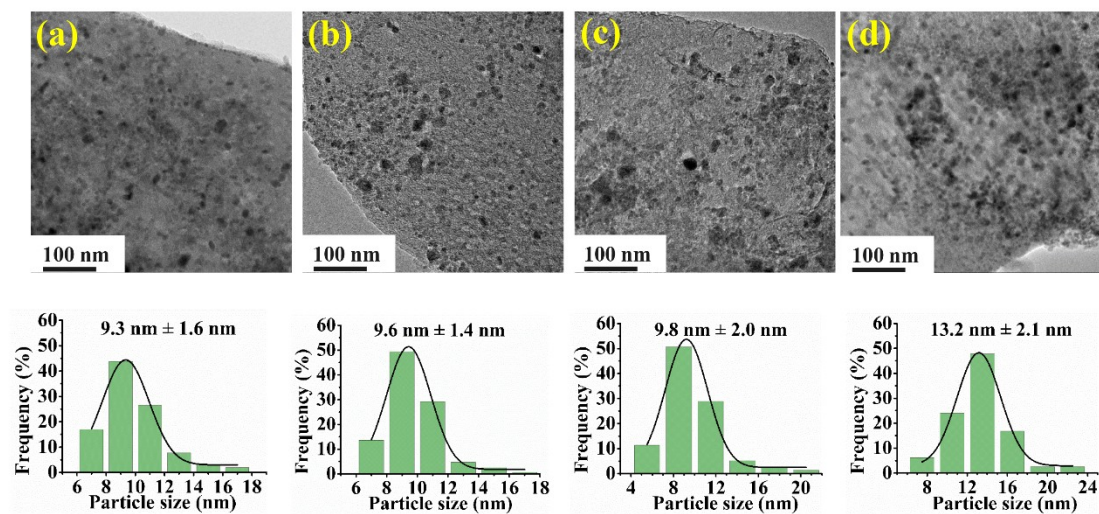


Fig. S5. TEM images of the used Fe/CNS(T) after 100 h reaction: Fe/CNS(900) (a), Fe/CNS(950) (b), Fe/CNS(1000) (c) and Fe/CNS(1100) (d) (diagrams below them are the iron particle size distribution of corresponding catalysts).

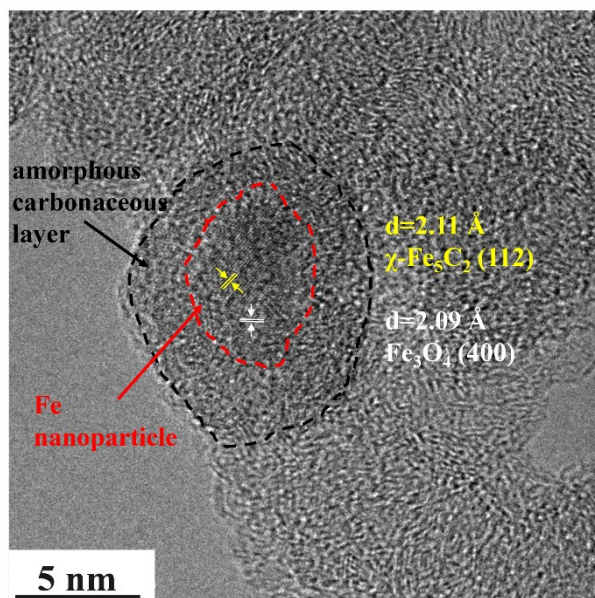


Fig. S6. HR-TEM images of the used Fe/CNS(1000).

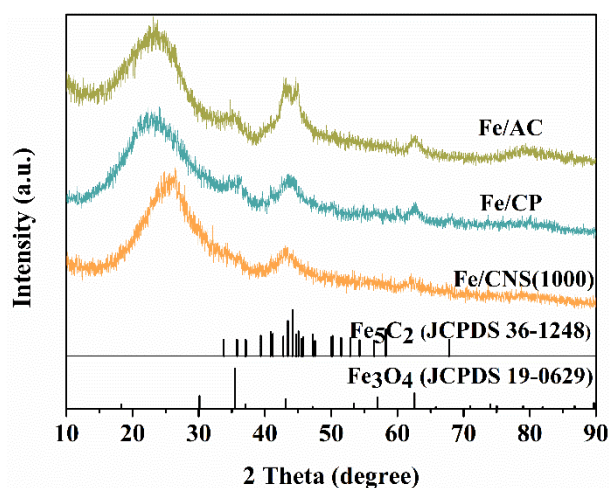


Fig. S7. XRD patterns of the used Fe/CNS(1000), Fe/CP, Fe/AC.

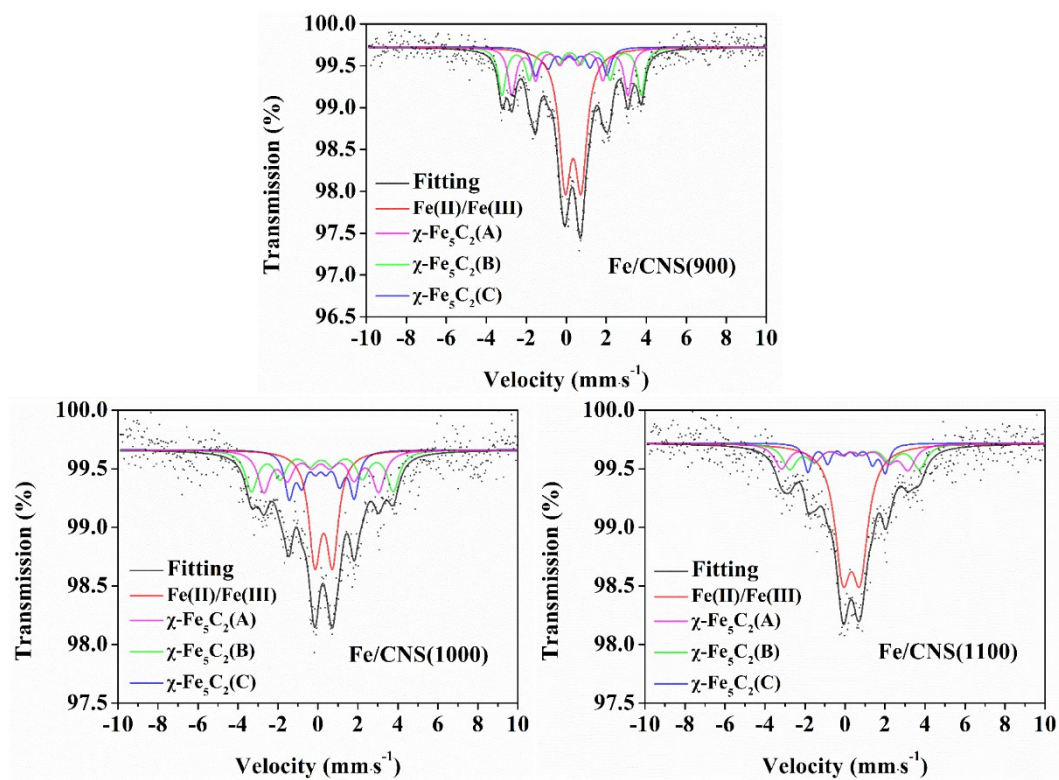


Fig. S8. ^{57}Fe Mössbauer spectra of the used Fe/CNS(900), Fe/CNS(1000) and Fe/CNS(1100) catalysts.